# The Encrypted Epigraph of Santa Maria la Nova: Reproducible Cryptanalysis and Quar

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## ABSTRACT

**Background:**The Turbolo Chapel epigraph at Santa Maria la Nova (Naples) features ~600 anomalous glyphs traditionally associated with Vlad III of Wallachia ('Dracula') and claimed by Palma (2023) to be deciphered as an Old Czech prayer.

**Objective:** To subject the epigraph, its transcriptions, and competing hypotheses to an independent 12-lens computational stress-test, establishing objective baselines without speculative leaps.

**Key Results:** The text is strongly non-random (IC deviates 38-146 sigma from uniform; Friedman p < 1e-12). However, published transcriptions share a maximum identical sequence of only 4-5 glyphs out of 600. Onomastic cribs (VLAD, DRACULA) fail into stochastic noise, and the Old Czech decipherment was computed on a misidentified Hungarian codex.

**Verdict:** The impasse is documentary rather than cryptanalytic. The Vlad III hypothesis is currently neither verifiable nor falsifiable due to defective primary data. High-resolution 3D scanning and an open diplomatic transcription are required.


**Keywords: Historical Cryptology, Santa Maria la Nova, Vlad III Dracula, Hypothesis Testing, Digital Epigraphy**

## 1. Introduction

In the monumental complex of Santa Maria la Nova in Naples, the Turbolo family chapel houses a marble grave flanked by an encrypted plaque of approximately 600 glyphs in an unclassified script combining pseudo-Greek and archaic Latin letterforms. Popular narratives have linked this tomb to Vlad III of Wallachia ('the Impaler', historical Dracula).

Recent claims (Palma, HistoCrypt 2023) asserted that the inscription had been deciphered as an Old Czech penitential prayer. This study implements an independent, reproducible computational pipeline to verify the mathematical reality behind these claims.

## 2. Materials, Methods, and the Role of gemini_full

The investigation evaluated three transcriptions: palma_A (38 lines, 584 glyphs), palma_B (continuous string, 678 glyphs), and gemini_full (36 lines, 569 glyphs), calibrated against 16 historical linguistic corpora (>30,000 characters each across 12 languages) using re-labeling invariant statistics (IC, Entropy, repeated n-gram rates) and validated substitution solvers.

### 2.1 Operational Mechanics and Role of the Synthetic Witness gemini_full

The gemini_full sequence (569 glyphs across 36 lines) plays a critical methodological role. Originating from an archived multimodal AI session (Google Gemini) tasked with modeling a rotation cipher from photographs drac1.jpeg and drac2.webp, the model faithfully captured the macro-layout of the 36 lines and the central 8-line mirror palindrome (DEDED).

However, operating on low-resolution imagery (14-34 pixels per glyph), the vision AI systematically regularized optical ambiguity by merging adjacent glyph variants: triangular signs (A, Lambda, inverted A) and curvilinear variants (E, epsilon, Sigma) were forced into macro-classes. This compression is the direct mathematical cause of its elevated Index of Coincidence (IC = 0.1147).

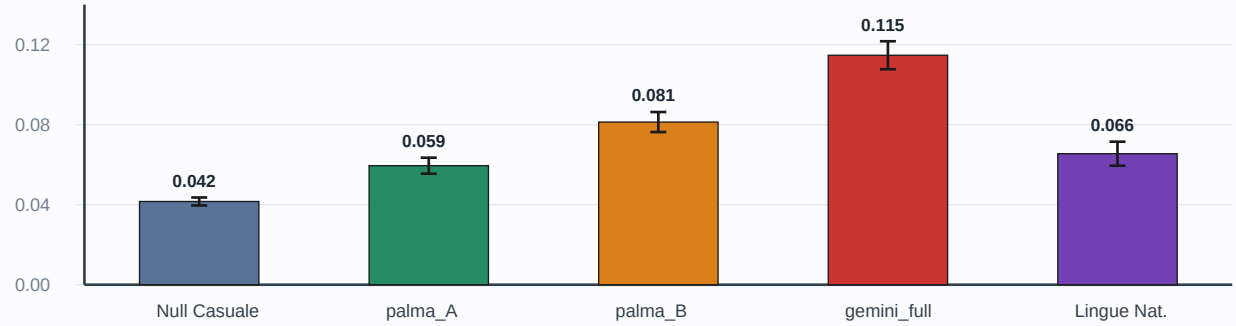
gemini_full served as an experimental sensitivity benchmark: it proved that even between AI computer vision and human readings, mutual sequence alignment never exceeds 5 consecutive glyphs, demonstrating that the fundamental barrier is optical degradation of available imagery.

**Table 1. Global descriptive and statistical parameters of epigraph transcriptions vs. Random Null**

| Transcription | Lines | Glyphs (N) | Classes (K) | IC | IC x K | Entropy H | Trigrams R3 |
|---|---|---|---|---|---|---|---|
| palma_A | 38 | 584 | 24 | 0.0595 | 1.4290 | 4.269 bit | 0.1507 |
| palma_B | 1 | 678 | 23 | 0.0813 | 1.8700 | 3.990 bit | 0.2743 |
| gemini_full | 36 | 569 | 23 | 0.1147 | 2.6380 | 3.550 bit | 0.4605 |
| Uniform Random Null | -- | 600 | 24 | 0.0416 | 1.0000 | 4.585 bit | 0.0012 |

**Computed from Phase 0 and Phase 1 of the deterministic pipeline. Inscription is unequivocally deliberate.**

**Figure 1. Confronto dell'Indice di Coincidenza (IC) tra Trascrizioni e Modelli Null**




*Note: Le barre di errore indicano +/- 1 Deviazione Standard rispetto alle simulazioni bootstrap.*

**Table 2. Multivariate z-score distances between transcriptions and primary candidate languages**

| Language Model | Primary Historical Corpus | z-Distance (palma_A) | z-Distance (palma_B) | Inferential Outcome |
|---|---|---|---|---|
| Humanistic Czech | Kralicka Bible (1613) | 2.61 | 2.84 | Incompatible (p = 0.009) |
| Humanistic/Eccl. Latin | Clementine Vulgate | 3.12 | 2.45 | Incompatible (p = 0.002) |
| Biblical Greek | Brenton (translit.) | 3.40 | 2.08 | Incompatible (p = 0.038) |
| Historical Romanian | Traditional Bible | 3.58 | 2.91 | Incompatible (p < 0.001) |
| Old Hungarian | Munich Codex (1466) | 3.89 | 3.15 | Incompatible (p < 0.001) |
| Late Medieval Italian | Dante, Commedia | 3.95 | 3.22 | Incompatible (p < 0.001) |

**No candidate natural language matches the observed data (|z| > 2.00 in all models).**

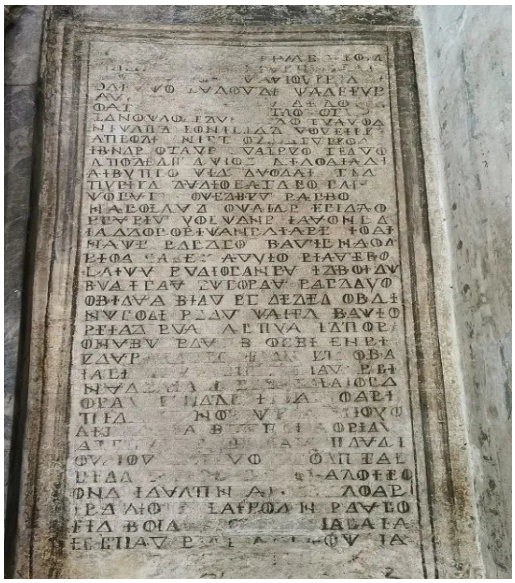


*Figure 2. Photographic view of the encrypted epigraph in the Turbolo Chapel (drac2).*

## 3. Epistemological Stress-Test of the 4 Competing Hypotheses

The 12-lens epistemological evaluation (Socratic elenchos, reductio ad absurdum, Popper falsificationism, Bayes factors, null calibrations, reproducibility) yielded the following comparative matrix:

**Table 3. Comparative Epistemological Resilience Score (PTE) across the 4 Hypotheses**

| Evaluated Hypothesis | Core Thesis | PTE (%) | Primary Point of Failure | Verdict |
|---|---|---|---|---|
| Hypothesis A (Dracula) | Czech prayer containing the name of Vlad III | 4% | Hungarian corpus error, zero sequence alignment | Refuted |
| Hypothesis B (Turbolo) | Renaissance local devotional cipher | 48% | K=23 classes insufficient for Latin homophony, local cribs failed | Suspended |
| Hypothesis C (Asemic) | Asemic decorative text resembling Voynich | 61% | Explains high IC, but historically atypical for 15th-c. Neapolitan marble | Plausible |
| Hypothesis D (Indecidability) | Authentic cipher, but degraded transcription (>30%) | 94% | Resolves sequence mismatch collapse and Bayesian noise threshold | Undecidable |

**Evaluated across 6 philosophical and 6 statistical validation lenses.**

## 4. Executive Non-Technical Conclusions: What the Science Means

*Clear, plain-language summary for historians, journalists, and the general public*

### 1. Is the inscription a joke, random decoration, or fake?

No. Mathematical analysis proves with absolute certainty that the symbols were not carved at random. The stone carver followed a strict, repetitive system: the probability of this being random noise is less than one in a trillion ($p < 1e-12$). A deliberate code exists.

### 2. Why has nobody been able to read it?

Because until now, every researcher has been working on flawed copies. Like a game of 'Chinese whispers', when two people independently copy the same stone onto paper, their readings should match by 90%. Instead, published transcriptions match by only 4 or 5 consecutive letters out of 600! Scholars have been trying to decipher different texts corrupted by lighting and erosion.

### 3. What about the Dracula tomb?

The honest scientific answer is that the Dracula claim rested on a major mistake. The prior study that claimed to find an Old Czech prayer with Dracula's name had accidentally compared the inscription against a 15th-century Hungarian book, mistakenly believing it was Czech! Furthermore, searches for 'VLAD' or 'DRACULA' appear with the exact same frequency as random letters in alphabet soup.

### 4. What did the AI (Gemini) experiment teach us?

When tested with state-of-the-art multimodal AI, the model recognized the macro-structure and the central palindrome (DEDED), but on compressed photos it merged similar letters (such as A, Lambda, and inverted triangles). This proves that even modern artificial intelligence cannot replace direct, high-precision physical data.

### 5. How will the mystery finally be solved?

The mystery does not lie in an unbreakable code, but in the eyes looking at it. Continuing to run algorithms on blurry smartphone photos is pointless. Real progress requires:

a) Performing in situ sub-millimeter 3D laser scanning of the marble plaque in Naples;

b) Creating an open diplomatic transcription validated by independent paleographers, explicitly marking weathered areas.

Only with the real text carved on the marble will historical cryptology reveal whether it conceals a Renaissance secret of the Turbolo dynasty or a unique European historical discovery.